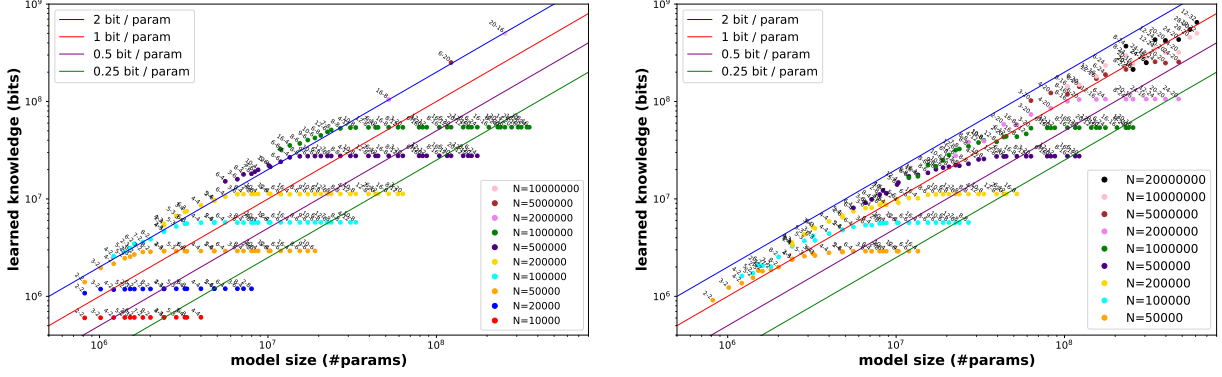




(a) bioS(N) data — **1000 exposures** — peak $R(F) \geq 2$ (b) bioS(N) data — **100 exposures** — peak $R(F) \geq 1$

Figure 1: Scaling laws for GPT2 pretrained on bioS(N) data using fp16 (mixed-precision) for 1000/100 exposures.

Conclusion. The *peak* capacity ratios consistently exceed $R(F) \geq 2$ (resp. ≥ 1) for 1000 exposures (resp. 100 exposures) of pretraining on each knowledge piece, **regardless of model depth/size**.

Remarks. Each dot ℓ - h represents GPT2 with ℓ layers, h heads, and $64d$ dimensions. The learned knowledge is calculated by the bit-complexity lower bound Theorem 3.2. The appendix also includes: Figure 11 showing similar results for bioS^{simple}(N) and bioR(N) data, Figure 14 demonstrating that the *same holds* for quantization using int8, Figure 10 confirming full extractability of all learned knowledge.¹³

Larger models? Training GPT2-20-16 on bioS(10M) for 1000 exposures costs 8.5 days with 64 A100s, while GPT2-12-32 on bioS(20M) for 100 exposures took 2.4 days. In our synthetic setting, we see no need to scale up further. Instead, we prefer to allocate GPUs to explore other aspects covered in this paper.

Remark 4.2. One must have $R(F) \leq R^{\max}(F)$, and equality is obtained if the model is *perfect*. For a fixed dataset, further increases in model size do not yield additional knowledge, thus $R^{\max}(F)$ approaches zero as the model size P increases. On the other hand, Theorem 3.2 implies, ignoring lower-order terms, that if the model parameters are 8-bit (such as int8), then $R(F) \leq 8$.

For our bioS(N) data, we define a slightly reduced capacity ratio by omitting the diversity term.¹¹

Definition 4.3. Given a model F with P parameters trained over the bioS(N) dataset $\mathcal{Z}$, suppose it gives $p_1 = \text{loss}_{\text{name}}(\mathcal{Z})$ and $p_2 = \text{loss}_{\text{value}}(\mathcal{Z})$, its *capacity ratio*¹²

$$R(F) \stackrel{\text{def}}{=} \frac{N \log_2 \frac{N_0}{e^{p_1}} + N \log_2 \frac{S_0}{e^{p_2}}}{P} \quad \text{and} \quad R^{\max}(F) \stackrel{\text{def}}{=} \frac{N \log_2 \frac{N_0}{N} + N \log_2 S_0}{P}$$

for $N_0 = 400 \times 400 \times 1000$ and $S_0 = 2 \times (12 \cdot 28 \cdot 200) \times 200 \times 300 \times 100 \times 263$ (c.f. Footnote 9).

Remark 4.4. Ignoring names, each person contains $\log_2(S_0) \approx 47.6$ bits of knowledge.

5 Base Scaling Laws

¹¹A version of Theorem 3.2 can be proven for this dataset with a simpler proof, as it excludes the diversity set. This could also mean the model has full prior knowledge of the diversity set (e.g., assuming a fixed set of 300 university names) without counting this knowledge towards its learned bits.

¹²Here, one can let $\mathcal{K} = \{\text{birth date, birth city, university, major, employer, gender}\}$ and accordingly define $\text{loss}_{\text{value}}(\mathcal{Z}) \stackrel{\text{def}}{=} \mathbb{E}_{n \in \mathcal{N}} \sum_{a \in \mathcal{K}} -\log \mathbf{Pr}_R[F^\perp(W(\mathcal{Z}), n, a, R) = v^*(n, a)]$.

¹³A distinction exists between memorizable knowledge (e.g., text memorized during pretraining) and knowledge flexibly extractable via instruction fine-tuning [3]; our results in this paper apply to both.

We first train a series of GPT2 models on the $\text{bioS}(N)$ datasets (see Section 2.2) using mixed-precision fp16. The training protocol ensures that each piece of knowledge is presented 1000 times, a process we refer to as “1000 exposures.” It’s important to clarify that this differs from making 1000 passes over the data. For example, a single pass through Wiki data might expose the knowledge (US, capital, Washington D.C.) 1000 times, whereas a pass through the Common Crawl might do so a million times. Our synthetic $\text{bioS}(N)$ data, trained for 1000 exposures, aims to replicate such scenarios.¹⁴ Our initial findings are as follows:¹⁵

Result 1 (Figure 1(a)). When trained for 1000 exposures on $\text{bioS}(N)$, with N ranging from 10K to 10M, GPT2 models with sizes from 1M to 0.5B parameters (*irrespective of depth or width*) demonstrate the following:

- (a) the peak capacity ratio $R(F)$ consistently exceeds $R(F) \geq 2$;
- (b) models with $R^{\max}(F) \leq 1.8$ attain near-perfect knowledge accuracies, i.e., $R^{\max}(F) \approx R(F)$;
- (c) across all models, $R(F) \leq 2.3$.

Remark 5.1. Result 1(a), 1(b), and 1(c) elucidate *three distinct facets* of the scaling law.

- Result 1(a) highlights the maximum capacity across models; however, this could be misleading if only a single model achieves this peak.
- Result 1(b) reinforces this by showing that all models with a maximum capacity $R^{\max}(F) \leq 1.8$ can achieve such maximum capacity, i.e., $R(F) \approx R^{\max}(F)$. In words, this indicates that for a dataset containing B bits of knowledge, selecting a model size $P \geq B/1.8$ is sufficient.
- Result 1(c) further strengthens this by indicating that no model exceeds capacity ratio 2.3.

For clarity, in subsequent results of this paper, we focus solely on the *peak* capacity ratio, with the understanding that observations similar to Result 1(b) and Result 1(c) **consistently apply**.

Knowledge extraction. The “2bit/param” result is not about word-by-word memorization. Even better, such knowledge is also flexibly extractable (e.g., via fine-tuning using QAs like “What is Anya Forger’s birthday?”) [3] and thus can be further manipulated in downstream tasks (such as comparing the birthdates of two people, or performing calculations on the retrieved knowledge, etc.) [4]. This is because our $\text{bioS}(N)$ data is knowledge-augmented: the English biographies have sufficient text diversities [3]. We also verify in Appendix A.2 that such knowledge is extractable.

5.1 Data Formats — Diversity and Rewriting

We conduct the same analysis on $\text{bioS}^{\text{simple}}$ and bioR . Recall from Section 2.2, $\text{bioS}^{\text{simple}}$ is a variant of bioS with reduced text diversity (one biography per person), while bioR is generated by LLaMA2, resulting in close-to-real human biographies. We have:

Result 2 (Figure 11 in Appendix A.3). In the same 1000-exposure setting, peak capacity ratios for GPT2 trained on $\text{bioS}^{\text{simple}}$ and bioR are also approximately 2, albeit slightly lower. Thus:

- Diverse data (rewriting the same data multiple times) does not hurt — and may sometimes improve — the model’s capacity!

¹⁴Within 1000 exposures, it’s likely that the same individual will have 1000 different biography paragraphs detailing the same knowledge (see Section 2.2). Therefore, 1000 exposures can occur within a single pass.

¹⁵We here focus on GPT2 models with depth ≥ 2 , and 1-layer models show slightly lower capacity ratios (see Figure 9). Our model selection covers most natural combinations of transformer width/depth, details in Appendix A.

Let’s highlight the significance of Result 2. Recall from Section 2.2:

- Training on $\text{bioS}^{\text{simple}}$ data for 1000 exposures equals 1000 passes over the data.
- Training on bioS data for 1000 exposures is less than 1 pass.
- Training on bioR data for 1000 exposures equals 25 passes.

Therefore, comparing bioS and $\text{bioS}^{\text{simple}}$, it’s more advantageous to rewrite the data 1000 times (in this ideal setting), training each for one pass (as done in the bioS data), rather than training the same data for 1000 passes (as done in the $\text{bioS}^{\text{simple}}$ data). This is because, without data diversity, the model wastes capacity memorizing sentence structures, resulting in a capacity loss.

In a realistic scenario, tools like LLaMA2 can rewrite pretrain data like we did in bioR . Rewriting data 40 times can produce 40 distinct English paragraphs, sometimes with (different) hallucinated contents. Does this require the model to be 40x larger? No, our comparison between bioS and bioR shows that, if trained for the same duration (40 rewrites each for 25 passes), the model’s capacity ratio remains nearly the same, slightly lower due to irrelevant data introduced by LLaMA2.

Allen-Zhu and Li [3] suggested that rewriting pretraining data is crucial for making knowledge extractable rather word-by-word memorization.¹⁶ However, they did not explore the impact on the model’s capacity. Our paper addresses this gap, indicating that rewriting pretraining data does not compromise — and may even enhance — the model’s knowledge capacity.

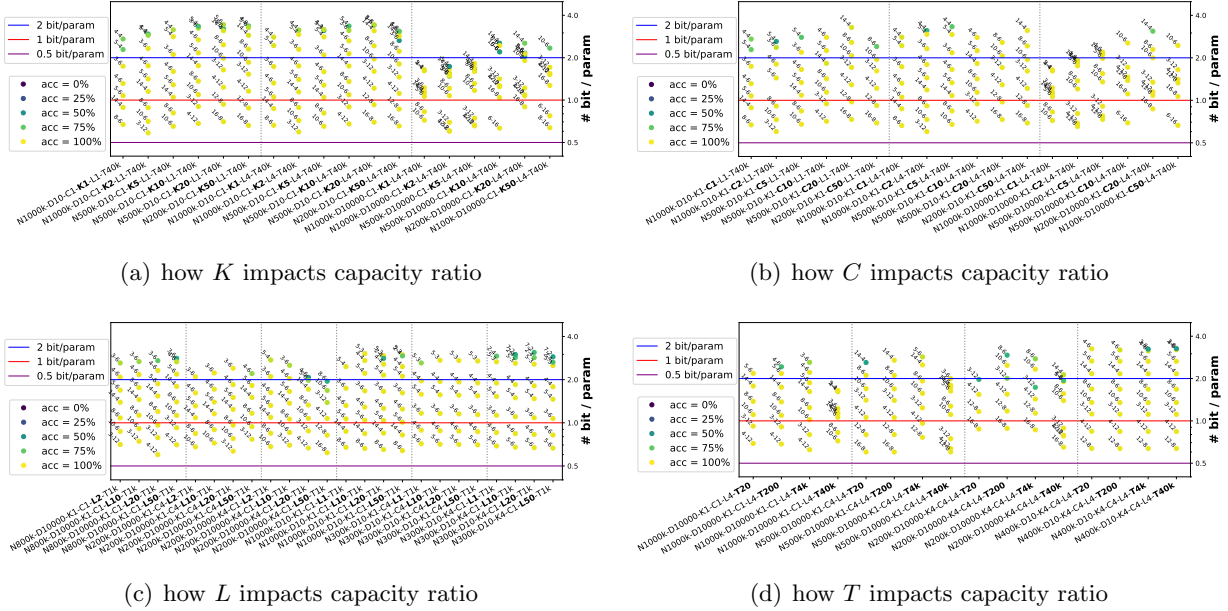


Figure 2: Scaling laws for GPT2 models trained on the $\text{bioD}(N, K, C, D, L, T)$ data for **1000 exposures**.

Conclusion. The *peak* capacity ratios consistently exceed $R(F) \geq 2$ with a wide range of K, C, D, L, T .

Remarks. Models with accuracies $\leq 50\%$ are excluded here but included in Figure 12. We disregard N ’s influence, akin to Figure 1(a), and concentrate on the five hyperparameters K, C, L, T, D . Each of the four sub-figures varies a primary hyperparameter while fixing the other four. More details in Appendix A.4.

¹⁶As demonstrated by [3], in low-diversity datasets like $\text{bioS}^{\text{simple}}$, knowledge can be word-by-word memorized but is nearly 0% extractable for downstream tasks. Others discover that rewriting data can improve the reversal extractability of knowledge [4, 11].

5.2 Parameterized Scaling Laws

We further investigate scaling laws within the $\text{bioD}(N, K, C, D, L, T)$ data family. Unlike with human biographies, where variation is limited to N , the bioD dataset allows for more flexible manipulation of the remaining hyperparameters K, C, D, L, T . This enables us to examine how variations in these parameters affect the model’s peak capacity.

Result 3 (Figure 2). Across a broad spectrum of values, with K, C ranging from 1 to 50, D from 10 to 10,000, L from 1 to 50, and T from 20 to 40,000, we observe that:

- GPT2 models consistently exhibit a peak capacity ratio $R(F) \geq 2$.

6 Training Time vs Scaling Law

What if the model is not sufficiently trained? For instance, there might be instances where knowledge appears only 100 times throughout the pretraining phase. We also calculate the capacity ratios for models trained with 100 exposures on $\text{bioS}(N)$. Our findings can be summarized as follows:

Result 4 (Figure 1(b)). When trained for only 100 exposures on the $\text{bioS}(N)$ dataset, with N ranging from 10K to 10M, across a broad spectrum of GPT2 models with sizes from 1M to 0.5B, the peak capacity ratio $R(F)$ consistently exceeds $R(F) \geq 1$.

Therefore, although 1000 exposures may be necessary for a model to reach its maximum storage capacity, training with just 100 exposures results in a capacity loss of no more than 2x.

In Section 10, we shall also consider knowledge that has *extremely low* (e.g., 1) or *high* (e.g., 1M) exposures. It may not be interesting to study them in isolation, but it becomes more intriguing when they are examined alongside “standard” knowledge, which has appeared, for instance, for 100 exposures, and how this impacts the model’s capacity. These will be our Result 10 through 12.

7 Model Architecture vs Scaling Law

Several transformer architectures have been widely adopted, with LLaMA and Mistral among the most notable. We outline their key distinctions from GPT2, with further details in Appendix B:

1. LLaMA/Mistral use so-called GatedMLP layers, which is $V(\sigma(W_1x) \cdot (W_2x))$ instead of $V\sigma(Wx)$. Shazeer [29] suggested that gated activation might yield marginally improved performance.
2. Unlike GPT2, LLaMA/Mistral do not tie weights.
3. Mistral features larger MLP layers compared to GPT2/LLaMA.
4. Mistral promotes group-query attention, not so by GPT2/LLaMA.
5. LLaMA/Mistral employ a different tokenizer than GPT2.
6. GPT2 uses the *gelu* activation function, LLaMA/Mistral opt for *silu*.
7. GPT2 implements layer normalization with a trainable bias.

Do these architectural variations impact the models’ maximum capacities? Our findings suggest that, in terms of knowledge capacity, GPT2 — when enhanced with rotary embedding and without dropout — performs no worse than any other architecture choice above in the sufficient training regime. We summarize the main findings below, deferring details to Appendix B.1: